

Attachment D. Expert Accuracy and Completeness Rubric

Readability of Online Patient Education Materials for Pre-Procedure Cardiac CT, With an Artificial Intelligence

Rewrite Arm and Lay Reader Assessment

Principal Investigator: Muhammad Naeem, MBBS, MD

Completed by a subspecialty cardiothoracic radiologist. The reviewer sees the original passage and one rewritten passage side by side.

Scoring instructions

The reviewer is blinded to which chatbot produced each rewrite; rewrites are stripped of any model identifier and presented in shuffled order. For each rewrite the reviewer scores the five items below. Every item uses a 1 to 5 scale on which 5 is best. Scores are recorded in the master data file (Attachment G) as expert_accuracy, expert_completeness, expert_added, expert_safety, and expert_overall.

Item (data field)	What it measures	Score 1 — worst	Score 3 — middle	Score 5 — best
Accuracy expert_accuracy	Factual correctness of the rewrite compared with the original passage.	Contains one or more serious factual errors.	A minor inaccuracy that would be unlikely to harm a patient.	Fully accurate; no factual errors.
Completeness expert_completeness	Whether important information from the original is retained.	Important information is missing.	Some non-essential detail is lost, but the key points are retained.	All important information is retained.
Added or unsupported content expert_added	Whether the rewrite introduces claims not supported by the original.	Adds one or more unsupported or invented medical claims.	Adds minor wording not in the original, but nothing misleading.	Adds nothing that is not supported by the original.
Safety and preparation information expert_safety	Whether safety, risk, and preparation instructions are preserved and correct.	A safety or preparation instruction is lost, made wrong, or made unsafe.	Safety information is present but less clear than in the original.	All safety and preparation information is preserved and correct.
Overall suitability expert_overall	Whether the rewrite could be given to a patient as written.	Not suitable; could mislead a patient.	Usable only after minor edits.	Suitable to give to a patient as written.

Direction of scoring. For every item a higher score is better, including expert_added, where a score of 5 means no unsupported content was added. If a downstream analysis instead reports added content as a

count on which a higher number is worse, expert_added is reverse-scored for that analysis and the direction is stated explicitly in the analysis code.

Free-text field

List any specific factual error, omission, or added claim, quoting the sentence at issue. This list becomes a results table in the manuscript.

Second reader

If a second subspecialty reader is available, a subset of at least 20 percent of rewrites will be double-scored and agreement reported.